Proposal: Real-time Data Warehouse for Detection & Anomaly Analysis

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1. Chosen Domain / Dataset

Domain: Financial Transactions / Fraud Detection in Digital Payments.

Dataset: *Metaverse Financial Transactions Dataset* (CSV).

- **Steps** → Simulation time (1 step = 1 hour, total 744 steps = 30 days).
- **Transaction Types:** CASH-IN, CASH-OUT, DEBIT, PAYMENT, TRANSFER.
- **Attributes:** Transaction amount, balances before/after, sender, receiver.
- **Fraud Label:** `isFraud` indicates whether a transaction is fraudulent.

This dataset is ideal for building a data warehouse for fraud detection and anomaly analysis because it provides rich **transaction-level data** and **fraud indicators**.

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2. Star Schema: Fact & Dimension Tables

We will design the warehouse using a **star schema**, where the central **Fact_Transaction** table records all transactions, and supporting dimensions provide context.

Fact Table: Fact_Transaction

Column	Description
transaction_id (PK)	Unique transaction identifier
datetime_id (FK)	Links to `Dim_DateTime`
sender_id (FK)	Links to `Dim_Customer`
receiver_id (FK)	Links to `Dim_Customer`
transaction_type_id (FK)	Links to `Dim_TransactionType`
amount	Transaction amount
oldbalanceOrg	Sender’s balance before transaction
newbalanceOrg	Sender’s balance after transaction
oldbalanceDest	Receiver’s balance before transaction
newbalanceDest	Receiver’s balance after transaction
isFraud	Binary flag (0 = normal, 1 = fraud)

Dimension Tables

`Dim_DateTime`

datetime_id (PK) step hour day month

(step = simulation hour, maps to real-world time)

`Dim_Customer`

| customer_id (PK) | customer_name | customer_type (Normal / Merchant) | region (if available) | risk_profile |

- Customers with IDs starting with **M** are merchants.

`Dim_TransactionType`

transaction_type_id (PK)	type_name	description
1	CASH-IN	Cash deposit into account
2	CASH-OUT	Withdrawal
3	DEBIT	Bank debit
4	PAYMENT	Merchant payment
5	TRANSFER	Peer-to-peer transfer

☑ This schema allows **flexible slicing** of transactions by time, customer, type, and fraud status.

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3. Planned ETL / ELT Workflow

1. **Extract**
   - Load raw CSV file into staging tables in PostgreSQL using pgAdmin4.
2. **Transform**
   - Clean missing values (esp. merchant balances).
   - Standardize transaction types (map to IDs).
   - Split datetime `step` into hour/day/month.
   - Identify and separate **merchants vs normal customers**.
   - Derive fraud metrics (e.g., fraud ratio per customer).
3. **Load**
   - Insert dimension records (`Dim_DateTime`, `Dim_Customer`, `Dim_TransactionType`).

    - Populate `Fact_Transaction` with foreign keys.
    - Add computed metrics (`fraud_flag`, aggregates).

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4. Plan for Analytics / BI Outputs

We will connect PostgreSQL DW to a BI tool (Power BI / Tableau) for visualization.

Dashboards & KPIs

- **Overview Dashboard**
    - Total transactions
    - Total transaction amount
    - Fraudulent transactions (count, %)
    - Average transaction value
- **Fraud Monitoring Dashboard**
    - Daily/hourly fraud trend
    - Fraud % by transaction type
    - Top risky customers (highest fraud ratio)
    - Top merchants receiving fraudulent funds
- **Transaction Type Analysis**
    - Distribution of transactions across types (CASH-IN, CASH-OUT, etc.)
    - Fraud likelihood by type (e.g., TRANSFER → CASH-OUT patterns)
- **Customer Insights**
    - Largest senders/receivers by volume
    - Accounts with balance inconsistencies
    - Fraud clusters (same sender multiple suspicious transactions)
- **KPI Cards**
    - Fraud-to-total ratio (%)
    - Total loss due to fraud (sum of fraudulent amounts)
    - Average anomaly score (if calculated)

☑ This project will demonstrate **end-to-end Data Warehousing**, covering:

- **Schema design (Star Schema)**
- **ETL/ELT pipeline**
- **Fraud analytics with BI dashboards**